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Chronocept: Instilling a Sense of Time in Machines

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TL;DR: The world's first benchmark to check if AI can understand time in linguistics.

"Humans know when facts expire. AI doesn't."

Prior approaches reduce time to static labels or binary transitions (valid/invalid), neglecting the gradual and asymmetric nature of temporal decay:

$$\text{validity}_i(t) = \begin{cases} \text{True} & \text{if information } i \text{ is valid at } t, \\ \text{False} & \text{otherwise} \end{cases} \quad (1)$$

We model temporal validity as a continuous-time marginal probability function:

$$p_i(t) \triangleq P(\text{validity}_i(t) = 1),\ p_i : T \to [0, 1] \quad (3)$$

Temporal Validity Prediction (TVP) as a Continuous Distribution

Mathematical Modelling

We define a binary random variable *validity$_i$(t)* in *{0,1}*, and model the instantaneous marginal probability of validity using a skew-normal distribution for a parameterised representation:

$$f(x; \xi, \omega, \alpha) = \frac{2}{\omega}\, \phi\left(\frac{x-\xi}{\omega}\right)\, \Phi\left(\alpha\, \frac{x-\xi}{\omega}\right)$$

- ξ (location parameter) determines the time at which an event is most likely valid,
- ω (scale parameter) governs the duration of validity, and
- α (shape parameter) controls the skewness (+ve values yield right skew and -ve values yield left skew)

No boundary conditions (such as *p(0) = 1* or monotonic decay) are assumed, allowing delayed onset, non-monotonic decay, and complex patterns.

The time axis is compressed using a monotonic log transformation with base 1.1 to avoid sparse coverage

MATRES-based Axes Decomposition

Each sample is split into eight semantically distinct axes, following the multi-axes annotation framework (MATRES) proposed by Ning et. al (2018).


```
{
    "_id": "H0028",
    "parent_text": "They are discussing a philosophical concept, whereas an online forum
simultaneously erupts in debate over similar ideas. They believe open dialogue fosters
clarity, yet they recognize tensions may escalate. They intend to document their
conclusions, hoping to contribute thoughtfully to the discussion."
    "axes": {
        "main_outcome_axis": "They are discussing a philosophical concept,",
        "intention_axis": "They intend to document their conclusions, hoping to
contribute thoughtfully to the discussion.",
        "opinion_axis": "They believe open dialogue fosters clarity,",
        "hypothesis_axis": "",
        "generic_axis": "",
        "negation_axis": "",
        "static_axis": "whereas an online forum simultaneously erupts in debate over
similar ideas. yet they recognize tensions may escalate.",
        "recurrent_axis": ""
    },
    "target_values": {
        "location": 39.865,
        "scale": 13.265,
        "skewness": 4.25
    }
}
```


Figure 1: Composition of samples in Chronocept benchmarks.

Benchmarks at a Glance

IAA Metric	BI	BII
ICC	0.843	0.893
Jaccard Index	0.624	0.731
P_k **Metric**	0.233	0.009

Table 1: IAA metrics for segmentation, axis categorization, and temporal validity annotation across both benchmarks. For P_k, lower is better, with values ranging from 0 (perfect agreement) to 1 (chance-level).

Our multi-stage annotation pipeline evaluates segmentation (P_k), axis categorization (Jaccard), and temporal validity (ICC) independently; only ICC is reported as benchmark-wide IAA.

Benchmark	Training	Validation	Test
Benchmark I	878	247	129
Benchmark II	365	104	55

Table 2: Dataset Composition and Splits.

Benchmark	Mean Length (μ)	SD (σ)
Benchmark I	16.41 tokens	1.56 tokens
Benchmark II	56.21 tokens	6.21 tokens

Table 4: Sentence Length Statistics for Benchmarks.

Parameter	Location (ξ)		Duration (ω)		Skewness (α)	
Benchmark	**Mean (μ)**	**SD (σ)**	**Mean (μ)**	**SD (σ)**	**Mean (μ)**	**SD (σ)**
Benchmark I	54.2803	20.4169	11.5474	3.7725	-0.0158	1.3858
Benchmark II	46.1511	13.3839	9.5553	2.5725	0.0275	1.1773

Table 5: Temporal Parameter Distribution Statistics for Benchmarks.

Results & Impact

Baseline Results

MT-DNN: best accuracy. SBERT-FFNN: best calibration. Also, Parameter-space Gaussian NLL outperforms label-space MSE on calibration.

Metric	MSE		MAE		R^2		NLL	
Baseline	BI	BII	BI	BII	BI	BII	BI	BII
DeBERTa-v3	695.7505	544.5860	14.8318	13.7358	-1.3404	-2.7730	9.9973	11.6107
Z DistilBERT	737.3157	640.2664	15.3509	14.7484	-1.5092	-3.0546	11.3089	13.6647
MT-DNN	**108.3768**	**64.5708**	**5.9425**	**4.5253**	**0.0314**	**-0.0183**	4.5117	4.1865
RoBERTa	909.5181	748.8589	17.2330	16.0097	-1.8687	-3.5666	15.2214	15.2253
SBERT-BiLSTM	171.7044	107.4771	8.5698	6.0472	-2.3193	-1.1621	4.4084	4.1373
SBERT-FFNN	155.8747	153.2178	7.9780	8.5744	-2.0203	-6.3268	**4.3769**	**3.9842**

Metric		RMSE			Spearman		
Baseline	**Benchmark**	ξ	ω	α	ξ	ω	α
DeBERTa-v3	Benchmark I	45.5250	3.6120	**1.2950**	-0.1179	0.0318	**0.1953**
	Benchmark II	40.2941	2.9834	1.1133	-0.2608	**0.3994**	0.0037
DistilBERT	Benchmark I	46.8732	**3.5750**	1.4375	-0.0545	0.0763	0.0860
	Benchmark II	43.7430	2.5137	1.0161	0.0754	0.2373	**0.2271**
MT-DNN	Benchmark I	**17.6030**	3.6852	1.2974	**0.5200**	0.1405	0.0115
	Benchmark II	**13.6704**	2.4017	1.0319	0.1807	-0.0488	0.0042
RoBERTa	Benchmark I	52.0884	3.6933	1.3079	-0.1243	-0.0371	0.0948
	Benchmark II	47.3300	**2.3293**	**1.0111**	0.0882	0.2670	0.1383
SBERT-BiLSTM	Benchmark I	20.7577	8.9916	1.8386	0.3710	0.1396	-0.0385
	Benchmark II	17.3793	4.3467	1.2238	0.1337	-0.2033	-0.0327
SBERT-FFNN	Benchmark I	19.6840	8.8119	1.5859	0.4687	**0.1670**	-0.0295
	Benchmark II	19.5139	8.4704	2.6672	**0.3690**	-0.0511	-0.1867

Ablations

- **Axis Removal**: MSE +5.95%, NLL +927% (BI)
- **Axis Shuffling**: NLL +358% (BII)
- **Loss Function**: Label-space MSE miscalibrates (NLL ~220-271); parameter-space Gaussian NLL is stable

Applications

Temporal-weighting in-retrieval · Knowledge Lifecycle Modelling · Fact Checking · Temporal QA · Semantic Signals for Proactivity

Future Work

Extending Chronocept to multimodal validity curves would capture phenomena with recurring. Document-level modeling and explicit atemporality labels remain open directions, as does scaling supervision beyond sentence-level scope to capture long-range temporal dependencies across discourse.